

Hostel code - 2613



Biosis

Intelligence Layer for the Biomass Economy

UNDERSTANDING THE PROBLEM STATEMENT

- India generates enormous quantities of agricultural and organic waste annually – much of which can be converted into renewable energy through Bio-CNG plants.
- Despite strong policy support and rising infrastructure investment, plant efficiency remains critically limited, leaving a massive energy and economic opportunity unrealized.

OPPORTUNITY

At ₹70/kg and 132 plants already operational, India's Bio-CNG sector represents a ₹23,506 crore annual market today – yet ₹2,351 crore is lost every year to operational inefficiency alone. At SATAT's 5,000-plant target, that loss scales to ₹5,250 crore annually across a ₹1,05,000 crore market.

Biosis is an AI-powered operational intelligence platform that optimizes biomass-based renewable energy infrastructure – starting with Bio-CNG plants. It delivers a data-driven decision layer enabling plant operators to optimize feedstock usage, stabilize digestion processes, and unlock carbon credit value.

Feedstock Type	Annual Availability
Agricultural residue	~150 million tonnes
Animal / livestock waste	~190 million tonnes
Municipal solid waste	~62 million tonnes
Sewage sludge	~50 million tonnes
Total	~450 million tonnes



FEEDSTOCK QUALITY UNCERTAINTY

- Biomass composition varies by crop type, season, storage, and moisture.
- Unknown biochemical inputs create **unpredictable digestion performance**.
- Current practice relies on **supplier declarations and visual inspection**.

SUBOPTIMAL FEEDSTOCK BLENDING

- Anaerobic digestion requires precise C:N ratio (25:1–30:1).
- Incorrect blending leads to **pH crashes**, **VFA buildup**, and **ammonia toxicity**.
- Decisions are manual, without predictive biochemical modeling.

REACTIVE PLANT OPERATIONS

- Problems detected only after **methane output drops**.
- Instability develops **days or weeks before detection**.
- Monitoring of pH, VFA, and gas composition is **manual and infrequent**.

IDENTIFYING THE PROBLEMS

UNMONETIZED CARBON CREDIT VALUE

- Plants qualify for credits under Verra VCS, Gold Standard, and India's CCTS.
- Potential revenue: **₹20–40 lakh per plant annually**.
- Most operators cannot maintain continuous **MRV documentation**, so credits go unclaimed.

DATA COLLECTED BUT NOT UTILIZED

- **SCADA systems** generate thousands of daily data points (temperature, pressure, gas flow, composition).
- Data is logged but **rarely analyzed for prediction or optimization**.
- Plants remain **data-rich but intelligence-poor**.



PESTEL ANALYSIS

POLITICAL

- India's SATAT initiative targets 5,000 Bio-CNG plants producing 15 million tonnes annually.
- OMC purchase obligations ensure guaranteed offtake, but operational technology gaps limit target achievement.

LEGAL

- India's Carbon Credit Trading Scheme (CCTS) and CBG agreements with OMCs incentivize efficient operation.
- Carbon credits require verified MRV documentation, which is currently manual and difficult for most operators.

ECONOMIC

- At full capacity, the Bio-CNG sector could generate ₹3+ lakh crore annually.
- Most plants operate at 60–75% capacity, creating an estimated ₹1,679 crore annual efficiency loss.

ENVIRONMENTAL

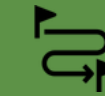
- Bio-CNG captures methane ($\approx 80\times$ stronger than CO_2 over 20 years) and converts it into renewable fuel.
- Plants running below capacity reduce both emission reductions and potential carbon credit value.

SOCIAL

- India burns ~ 92 million tonnes of crop residue annually, causing severe air pollution.
- Efficient biomass plants create stable demand for agricultural waste, providing farmers an alternative to burning.

TECHNOLOGICAL

- Anaerobic digestion requires precise control of feedstock composition, pH, and microbial balance.
- Most plants rely on manual monitoring despite collecting SCADA data, leaving a critical intelligence gap.



STRATEGIC SOLUTIONS OVERVIEW

Methane Yield Forecast Dashboard

- Rolling 7-day methane production forecast, updated daily with confidence intervals
- Powered by live **feedstock inventory**, **digester health**, and **blending recommendation** inputs
- Enables proactive procurement planning and accurate offtake commitments to **OMC partners**

Carbon MRV Automation

- Tracks **daily methane emission** reductions and generates monthly registry-ready reports (**AMS-III.D / VM0041**)
- Maintains a **complete audit trail** for seamless third-party verification
- Unlocks **₹20–40 lakh/year** in carbon revenue for a **100 TPD plant**—often making Biosis self-funding

Feedstock Intelligence Engine

- Assigns a **Feedstock Quality Score (0–100)**, flagging inhibitory compounds and **quality degradation risks**
- Predicts **biogas yield per tonne** before digestion begins
- Recommends optimal **utilization windows**, **eliminating procurement guesswork**

Predictive Blending Engine

- Solves **multi-variable optimization** to recommend exact daily blend ratios, maintaining **C:N (25:1–30:1)** and preventing **VFA buildup**
- Delivers simple, actionable output: "**Feed 40% paddy straw, 35% cattle dung, 25% press mud today**"
- Continuously **self-calibrates per plant**, improving accuracy with **every operational cycle**

Anomaly Detection & Plant Stability

- Ingests live **SCADA streams**. Three-tier alert system:
- **Watch** — early deviation detected
- **Caution** — corrective action recommended
- **Critical** — immediate intervention protocol

PROTOTYPE-BIOSIS OPERATOR VIEW

An integrated intelligence layer that synchronizes **Feedstock Intel** for intake optimization, **Digester Health** for real-time risk mitigation, and **Carbon MRV** for automated credit tracking across a high-performance **Plant Network**.

Live Methane Yield: Real-time m^3/kg tracking against rolling averages for performance transparency.

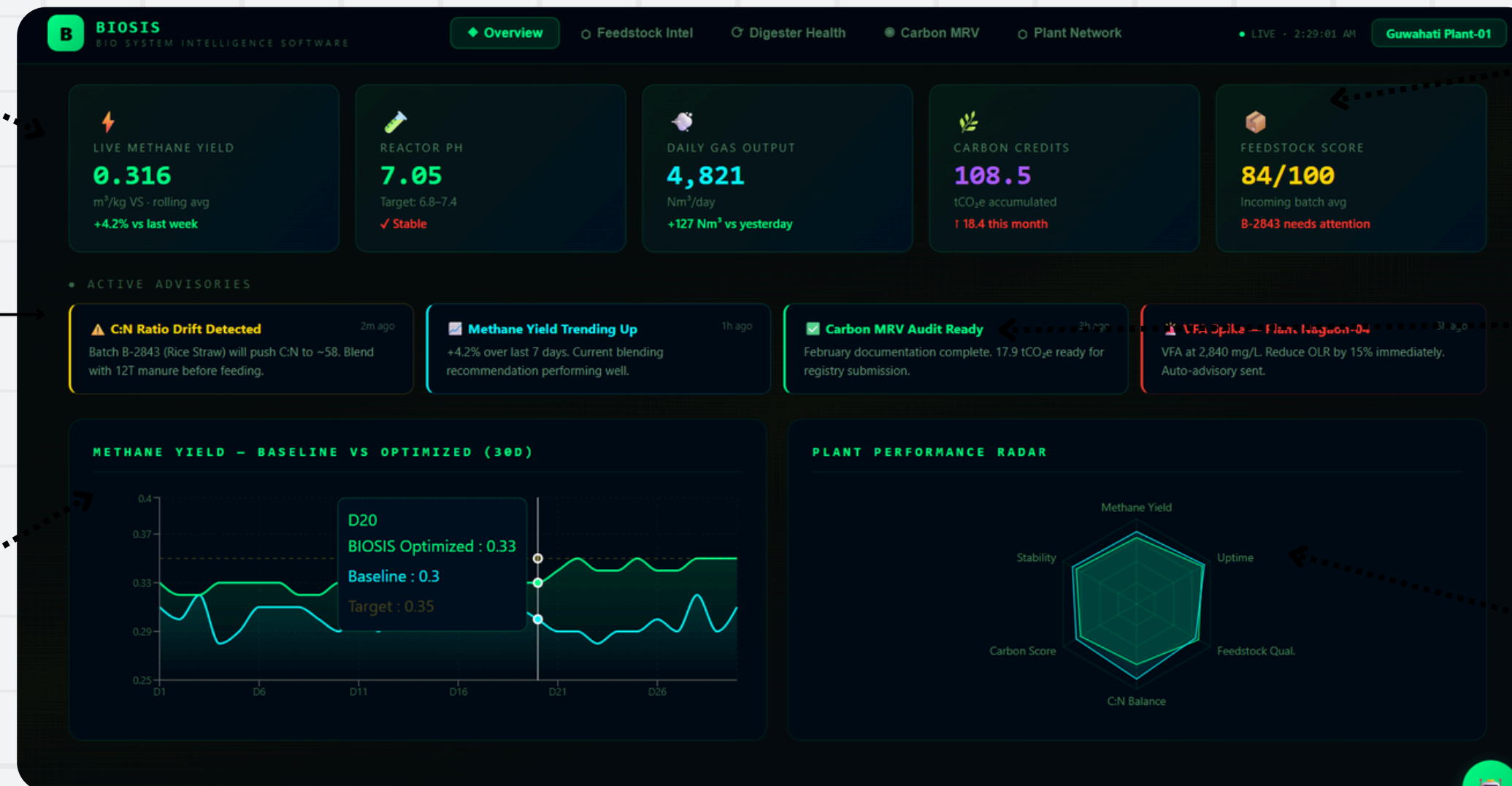
AI Advisories: Automated corrective actions for C:N drifts, VFA spikes, and organic overloads.

Predictive Analytics: 30-day "Baseline vs. Optimized" yield visualization and 7-day look-ahead forecasting.

Feedstock Intelligence: 0–100 quality scoring for incoming batches to prevent digester shock.

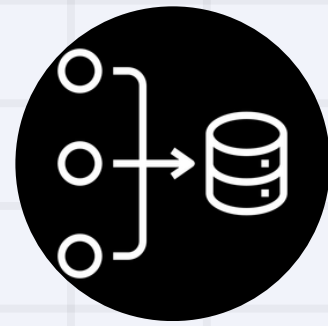
Carbon MRV: Audit-ready CO₂ tracking for seamless registry submission and credit generation.

Plant Performance Radar: 360° visual of stability, uptime, carbon score, and feedstock quality.



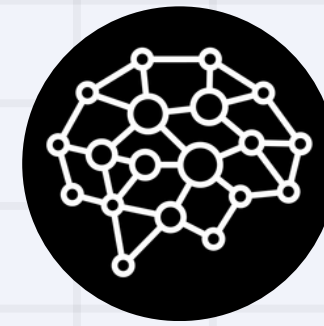
[Click here to see the Prototype](#)

TECHNOLOGICAL IMPLEMENTATION



Data Inputs

- **SCADA plant data:** real-time operational parameters (temperature, pH, gas flow)
- **Feedstock logs:** daily biomass quantity and composition records
- **Supplier profiles:** historical quality and reliability data
- **Moisture readings:** on-site measurements for feedstock variability
- **Lab calibration samples:** reference data to improve model accuracy



AI Models

- **Yield Prediction:** estimates methane output from different feedstocks
- **Blending Optimization:** recommends optimal feedstock mix for stable digestion
- **Failure Prediction:** detects early signs of biological or equipment issues

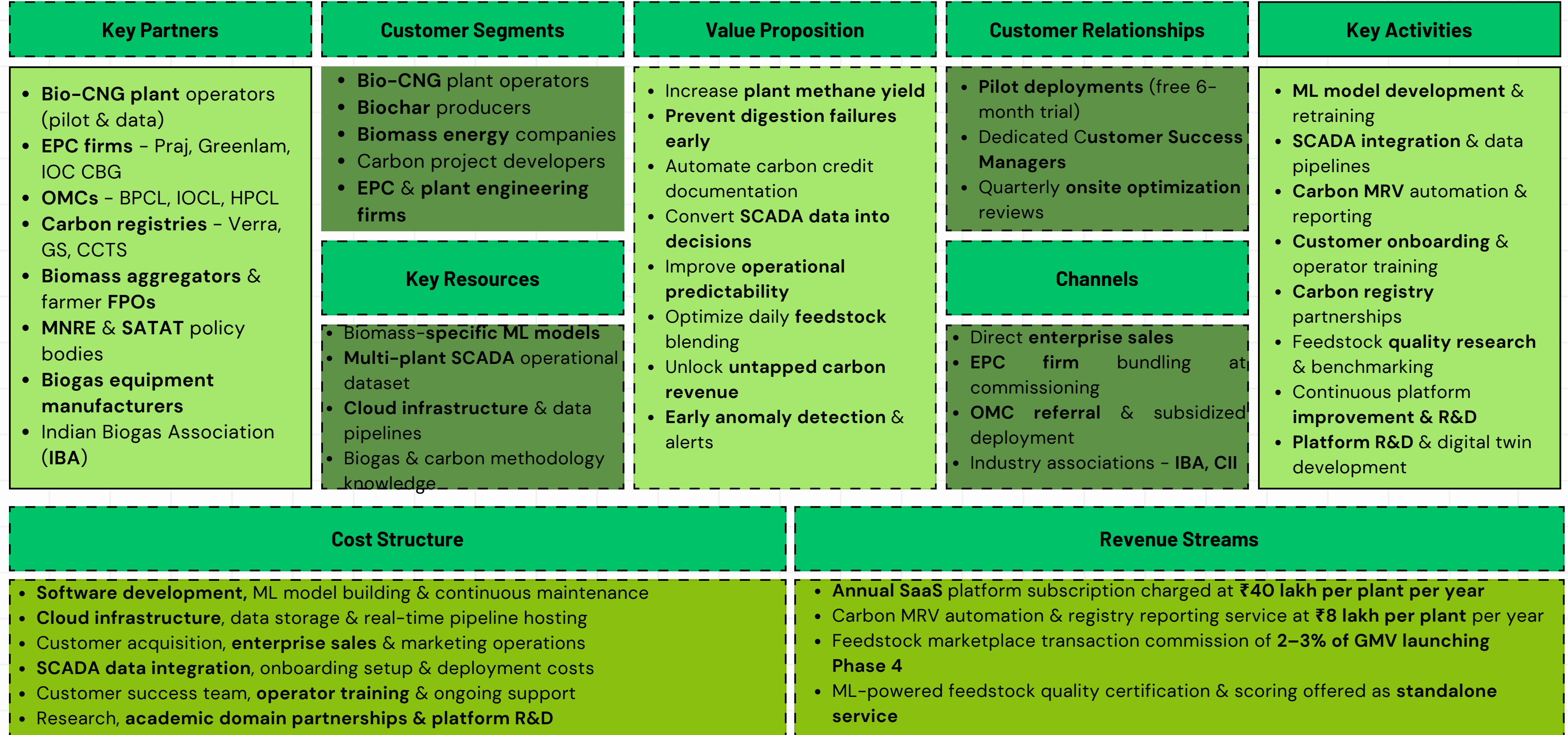


Platform Architecture

- **Cloud-based SaaS platform** for centralized data processing
- **API integrations** with existing SCADA and plant software
- **Operator dashboard** for real-time insights and recommendations
- **Carbon automation** engine for MRV reporting and credit tracking



BUSINESS MODEL CANVAS



MARKET SIZING

TAM Calculation

Basis: Global biogas and Bio-CNG SaaS + Carbon MRV platform potential per IEA, World Biogas Association, and IRENA estimates.

- **Total global biogas plants (operational + planned by 2030):** ~60,000 plants across Europe, China, India, Southeast Asia, and Americas
- **Average plant size (global blended):** 150–200 TPD equivalent; significantly larger than Indian SATAT plants
- **Global SaaS + MRV addressable fee:** ₹80 lakh/plant/year (blended; European/US plants priced at 3–5× India given plant scale and carbon market depth)
- **Basis for fee:** Process optimization SaaS captures <2% of incremental revenue generated; carbon MRV priced at ~10% of credit value per plant

TAM = 60,000 plants × ₹80 lakh = ₹4,80,000 crore (~₹4.8 lakh crore)

SAM Calculation

Basis: SATAT targets 5,000 plants by 2028–29; LOIs issued for 1,600+; realistic 5-year operational buildout pegged at 40% of policy target based on India's historical infrastructure execution rate.

- **Addressable plants (5-year India horizon):** 2,000 plants
- **SaaS fee:** ₹40 lakh/plant/year (basis: 5% yield uplift on 100 TPD plant = ₹9.1 crore incremental revenue; fee = <5% of value created)
- **Carbon MRV service:** ₹8 lakh/plant/year (basis: ~10% of voluntary carbon credit revenue per plant at ₹1,000/tonne × ~800 tCO₂/year)

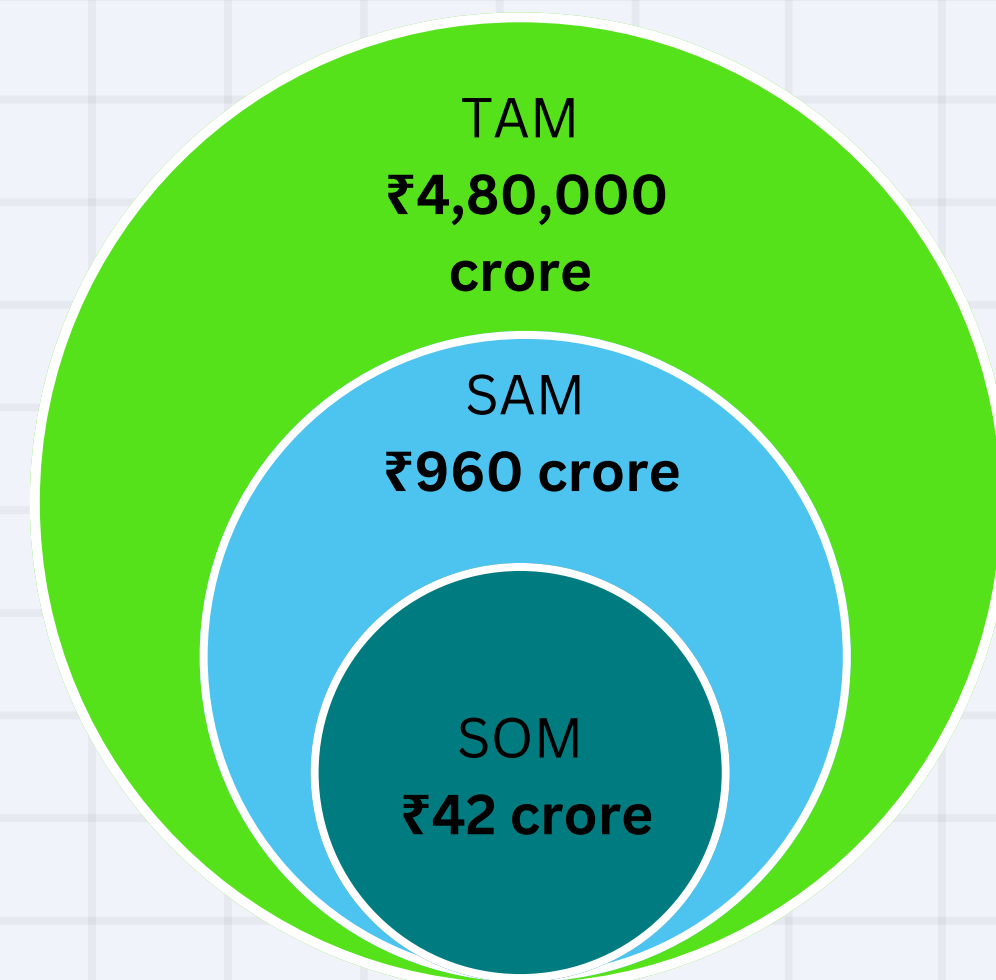
SAM = 2,000 × ₹48 lakh = ₹960 crore/year

SOM Calculation

Basis: Focused penetration of operational SATAT-registered plants above 30 TPD with active OMC offtake agreements — the niche where Biosis ROI is immediate, payback period is <12 months, and competitive alternatives are near-zero.

- **Target plants (30+ TPD, operational, OMC offtake):** ~200 qualifying plants by Year 3
- **Win rate:** 40% penetration of qualifying niche (justified by unambiguous ROI, SATAT compliance reporting as forcing function, lack of India-specific Bio-CNG SaaS competitors)
- **SaaS revenue:** 80 plants × ₹40 lakh = ₹32 crore
- **Carbon MRV services:** 80 plants × ₹8 lakh = ₹6.4 crore
- **Implementation & onboarding (one-time, Year 3 cohort):** 80 plants × ₹5 lakh = ₹4 crore

SOM (Year 3) = ~₹42 crore ARR (₹38.4 crore recurring + ₹4 crore onboarding)



THE ECONOMIC OPPORTUNITY

Layer 1 — What the Market Is Worth Today

Category	Operational Metric
Operational plants	132 plants across India
Combined daily capacity	920 tonnes/day
Annual production potential	3,35,800 tonnes
Bio-CNG price (OMC rate)	₹70/kg
Annual industry value	₹23,506 crore

Layer 2 — What Is Being Lost Right Now

Production Timeline	Capacity (Tonnes)
Daily Production	920
Monthly Production (Est.)	27,983
Annual Production	3,35,800

Layer 3 — What the Full Opportunity Looks Like

Market Value Metric	Value (INR)
Price per kg	₹70
Price per Tonne	₹70,000
Total Annual Market Value	₹23,506 Crore

Layer 4 — What Biosis Returns to a Single Plant

Metric	Value
Yield improvement	5% (conservative)
Additional gas revenue	₹70.3 lakh/year
Carbon credit uplift (MRV enabled)	₹20–25 lakh/year
Total value created	₹90–95 lakh/year
Platform cost	₹48 lakh/year
Net benefit	₹42–47 lakh/year
Payback period	< 4 months

The Multiplier Effect

*At **500 plant** deployments, Biosis unlocks **₹4,500+ crore** in additional industry value annually – while the platform itself generates **₹255 crore ARR**. Every rupee spent on Biosis returns **~18×** in value to the sector.*



COMPETITIVE LANDSCAPE

Parameter	 SenseHawk				
Primary Focus	Solar Monitoring	Industrial AI	Waste Management	Biogas Engineering	Biomass Intelligence
Feedstock Analytics	X	X	Limited	X	ML Scoring
Predictive Blending	X	X	X	X	Optimization Engine
Plant Optimization	Limited	Generic AI	Consulting	Limited	Continuous Optimization
Anomaly Detection	X	Generic	X	X	Biomass AI
Carbon MRV	X	X	Limited	X	Automated MRV
Biomass Specialization	X	X	Partial	Engineering	End-to-End
India Market Focus	✓	Limited	Limited	X	India-First
SaaS Platform	✓	✓	X	X	SaaS Model
Feedstock Marketplace	X	X	X	X	Future Phase

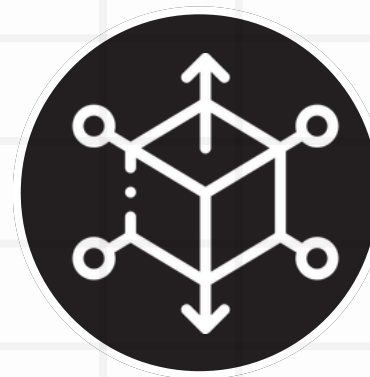
GO-TO-MARKET

PROOF-OF-ROI FIRST: SCALING THROUGH TRUST & PARTNERSHIPS



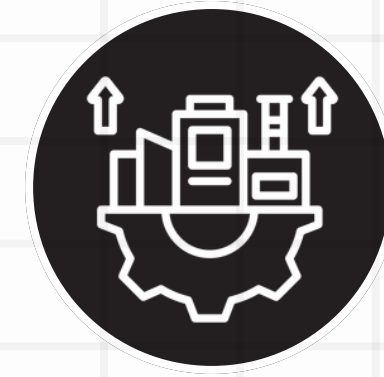
Phase 1: Proof of Value (Months 1–12)

- Target: 3–5 operational reference sites (50–150 TPD) in high-feedstock clusters (Punjab, Haryana, UP, MH, KA).
- Acquisition: Direct outreach (SATAT registry), warm intros via equipment OEMs (Praj, Greenlam), and a "6-month free pilot" offer in exchange for data/case studies.
- Success KPIs: %methane yield improvement, reduced instability, and registry-ready MRV documentation.



Phase 2: Commercial Launch & Scaling (Months 12–24)

- Target: 20–30 paying plants; 80% pilot-to-paid conversion via 3-year stabilized ARR contracts.
- **Strategic Growth Engines:**
- EPC Partnerships: Bundle Biosis into new plant commissions (Praj, Greenlam, IOC) with 15–20% revenue share.
- OMC Partnerships: Engage BPCL, IOCL, HPCL for subsidized deployment tied to supply volume.
- Industry Authority: Leverage IBA technical committees and CII/FICCI renewable forums.



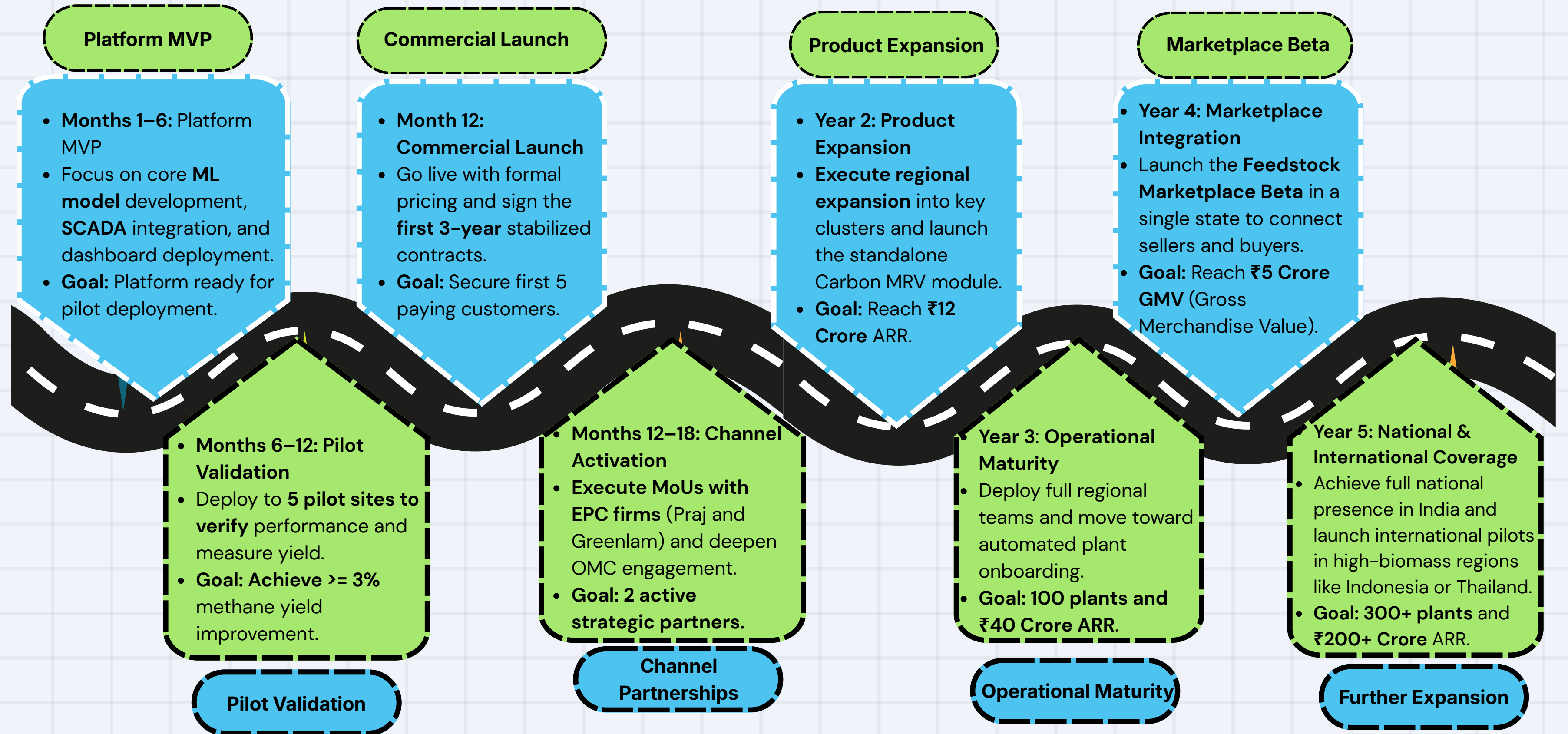
Phase 3 & 4: Expansion & Infrastructure (Year 2–5)

- Regional Clusters: Expanding into UP-Bihar (biomass potential) and Gujarat-Rajasthan (industrial/municipal waste).
- Carbon MRV: Standalone product launch (₹8 lakh/year) for plants needing carbon documentation.
- Marketplace Vision: Transform into the "Infrastructure Layer" of India's biomass economy, connecting farmers/aggregators to plants (2–3% transaction fee).



IMPLEMENTATION ROADMAP

Objective: From MVP to 300+ plants and ₹200+ Cr ARR.



FEASIBILITY & VIABILITY

Technical Feasibility

Infrastructure already exists

- Bio-CNG plants >30 TPD have SCADA (**temp sensors, gas flow meters, pressure & biogas analyzers**); integration via Modbus/OPC-UA software API, zero new hardware

Proven technologies

- XGBoost/LSTM models & **time-series anomaly detection** are commercially validated in industrial engineering; Biosis applies them domain-specifically to biomass

Training data ready

- **pre-trained** on IEA Bioenergy, DBFZ Germany & Biogas Research Center Sweden datasets + synthetic data from validated AD biochemical models; fine-tuned per plant on historical SCADA during onboarding

Operational Feasibility

Rapid Deployment

- **Software-only integration** enables full system setup within ~3–4 weeks, with no plant shutdown or operational disruption.

Minimal Operational Change

- Runs parallel to **existing plant management systems**, allowing operators to validate recommendations before full adoption.

Operator-Friendly Design

- Insights delivered in clear, **plain-language guidance** rather than **complex analytics**, ensuring adoption without requiring data science expertise.

Economic Feasibility

Low cost, fast ROI

- SaaS subscription model with zero capex; a plant recovers the annual subscription cost within **1–2 months at just 3% efficiency gain**

Reduces costly

- **Predictive anomaly detection** prevents process failures **costing ₹5–15 lakh per incident**; directly protecting operator margins

Improves bankability

- **Consistent**, predictable gas output strengthens offtake certainty, **improving plant IRR** and access to debt financing at scale



IMPACT

Energy

750,000 additional tonnes of renewable gas per year
— equivalent to fueling
~150,000 buses annually

Environmental & Social

Efficient, predictable plants create reliable agricultural waste demand — directly reducing the economic incentive for stubble burning across the Indo-Gangetic plain

Climate

1.8–2.25 million tonnes of additional CO₂e reduction per year — equivalent to removing 400,000 cars from the road.

Economic

₹3,500+ crore in additional industry value annually at 500 plant deployments — strengthening viability of India's entire biomass sector

Market & Deployment Scale

At 500 plant deployments, Biosis captures only 10% of SATAT's addressable market — leaving a 10x growth runway as India scales toward its 2030 biomass targets



FUTURE PLANS

Phase 1 — Advanced Plant Intelligence (Year 0–2)

Digital Twin: Real-time plant simulation for feedstock testing, predictive maintenance, and process optimization.

30-Day Yield Forecasts: Improves procurement and offtake planning.

Automated Interventions: System-guided corrective actions for rapid operator response.

Phase 2 — Carbon Market Integration and Adjacent Segments (Year 3–4)

Carbon Registry APIs: Integration with **Verra, Gold Standard, and India CCTS** for automated MRV and credit issuance.

Biochar Optimization: Balance biogas vs biochar yield using carbon price signals.

Biomass Certification: ML-based feedstock quality scoring for transparent procurement.

Phase 3 — AI-Driven Supply Forecasting and Marketplace (Year 4–5)

Supply Forecasting: 30–90 day biomass availability predictions using satellite, crop, and weather data.

Feedstock Marketplace: Platform matching sellers and plants; **2–3% transaction fee.**

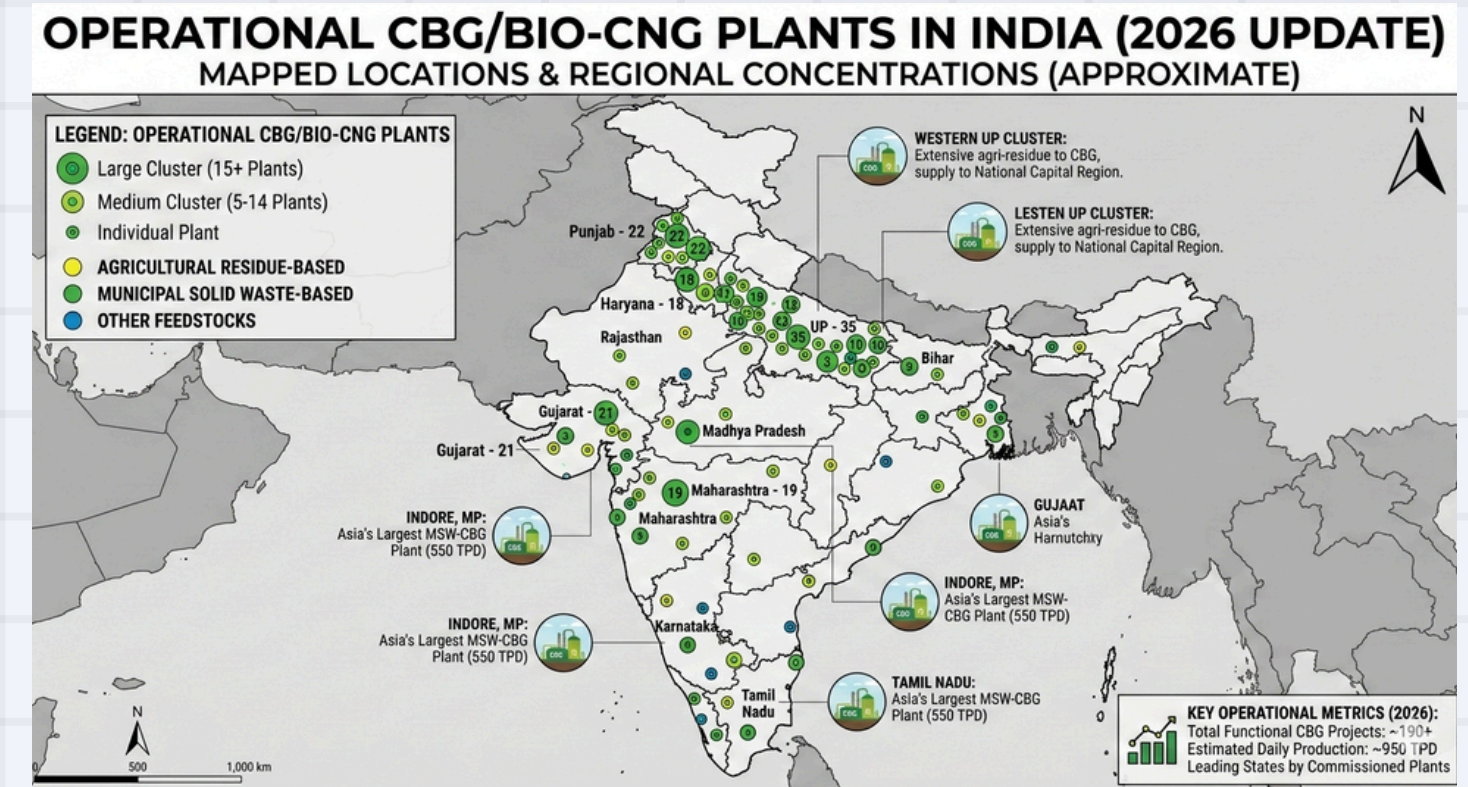
Global Expansion: Replicate model in **Indonesia, Thailand, Kenya, and Nigeria** (pilot Year 5).

Future Plans & Strategy
detailed report



APPENDIX

1. <https://www.petroleum.nic.in/satat>
2. <https://mnre.gov.in/bio-energy/current-status>
3. <https://www.niti.gov.in/sites/default/files/2022-07/Bioeconomy-report.pdf>
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THANK YOU